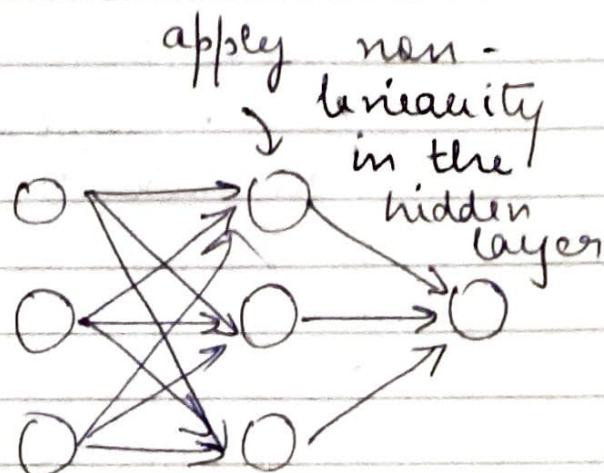


Speech Tech - Week - 8

18.1

FFNN - 1

Features	class
1.4 2.7 1.9	0
⋮ ⋮ ⋮	0
	1
	0



we can control the wts.

so, we have to train the neural network.

- initialise with random wts.
- we take first input & present a training pattern
- get some result, compare with target output
- adjust the wts. based on losses
- repeat this many times, and do this until convergence.
- finally, post training it shows a good decision boundary.
- wt-learning algs are dumb.

$$\boxed{MSE = \frac{1}{2} \sum_{i=1}^K (y_i - \hat{y}_i)^2} \rightarrow \text{mean sq. error}$$

$$\boxed{X_{ent} = - \sum_{i=1}^K y_i \log(\hat{y}_i)} \rightarrow \text{cross entropy loss}$$

$\hat{y}_i \rightarrow$ predicted value
 $y_i \rightarrow$ true value.

- these loss func^{ns} are used to evaluate the model.

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}} \quad \rightarrow \text{more helpful in probabilistic situa}^{\text{ns}}$$

this is discriminative

z_1 z_1
 $\vdots$ $\vdots$
 z_n z_k

$l = 1, 2, \dots, k$

finally apply softmax here.

$e^{z_i} \rightarrow$ goes from 0 to ∞ , can't be -ve.

$$0 < y_i = \frac{e^{z_i}}{e^{z_1} + \dots + e^{z_k}} = 1$$

L8.2 Part-2

- In order to reduce the error, the parameter wt. has to be changed in the direcⁿ of decreasing slope, i.e., the -ve of the denominator of the error funcⁿ

gradient descent method



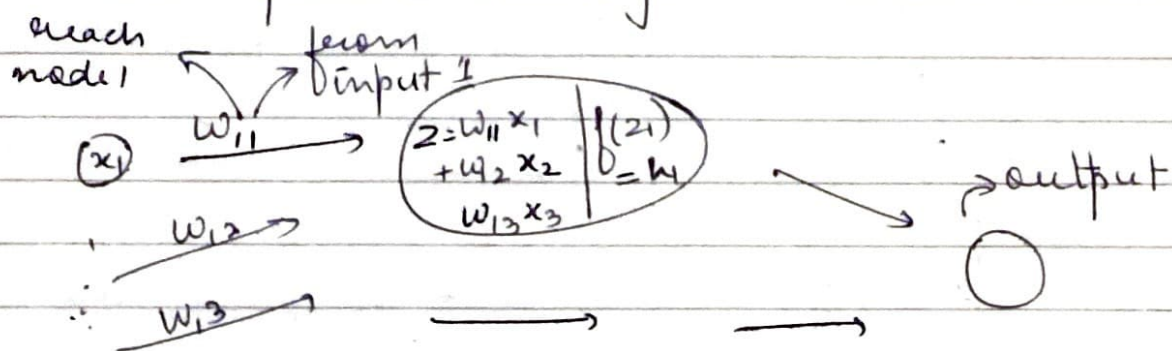
slope -ve $\Rightarrow$ increase the wt.
 slope +ve $\Rightarrow$ decrease the wt.

$$W_{\text{new}} = W_{\text{old}} - \eta \frac{dE}{dW}$$

error funcⁿ
(MSE/Xent)

learning rate

- $O_k \rightarrow$ unit k by sigmoid
- taking derivative of error, we apply chain rule.
- diff. btw. local & global minima.
 - $\hookrightarrow$ with large amt. of data, we can reach good minima during pre-training
 - DNN (deep neural network) $\rightarrow$ with multiple hidden layers.



$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\boxed{z = Wx}$$

L8.3 Rev. of FFNN

- neural networks are good, they provide non-linearity.

- softmax is discriminative (loss) training
- DNN helps to model more complex layers

LS.4 NN for speech

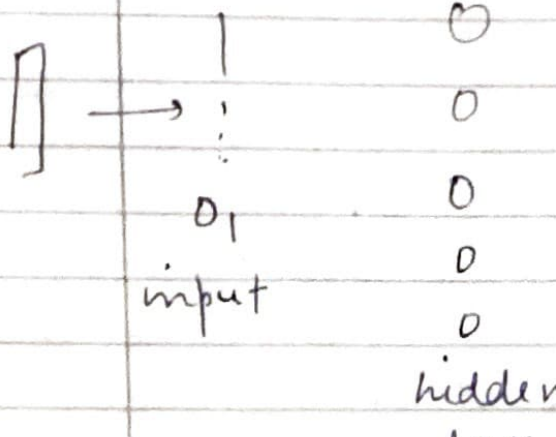


O_1 O_2
MFCC features

- in speech, we want to find out what sound each MFCC vector corresponds to? I presently is hidden from me.

- because of speaker variations, envt. noise for each phoneme → O_i MFCC vector

- GMM → DNN
 gets replaced by
 but after that GMM
- exhibit a lot of variability
- ∴, hybrid model

- $\rightarrow$ (prior)
- what GMM $\rightarrow 0, \rightarrow$ "classified" problem
- 

input

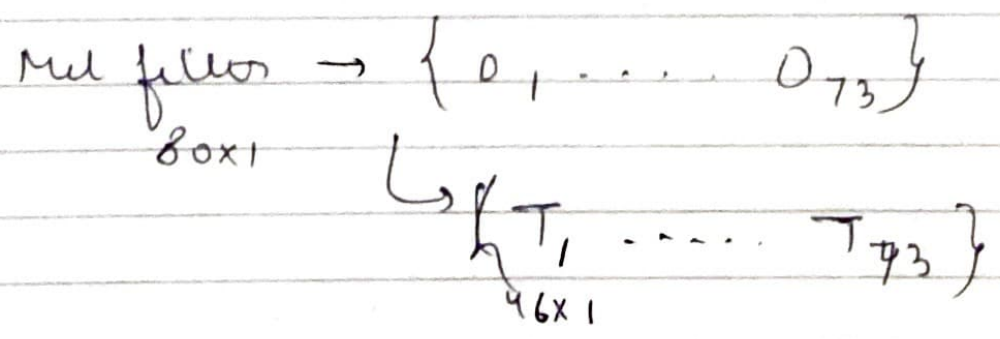
hidden layer

softmax to assign the probability

which of the following 45 phonemes class it belongs to (posterior)
- for every mel filter $\rightarrow$ i need to know sound class during training

L8.5 Intro to CTC -1

- alignment problem $\rightarrow$ for 25 ms which phoneme it corresponds to.



Alignment $\rightarrow$ flat start $\rightarrow$ uniformly segment

$\hookrightarrow$ use some prior related model (forced alignment)

neuroform $\rightarrow O_1 \dots O_T$

lexicon
b

use the model to align O vectors with the phonemes

- very problematic to correctly assign every frame to the corresponding phoneme (forced alignment)

- this is the idea of CTC

y ch ch ch s $\Rightarrow$ y ch s

y y ch ch s $\Rightarrow$ y ch s

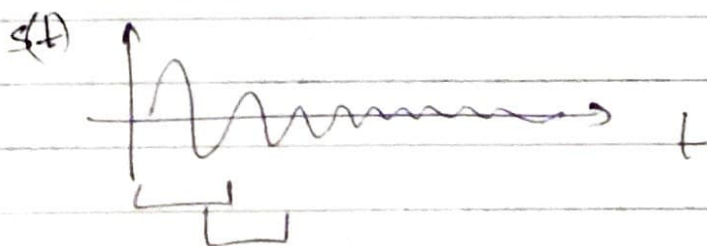
$\Downarrow$
lexicon
 $\Downarrow$
yes

don't be picky to map every target frame correct.

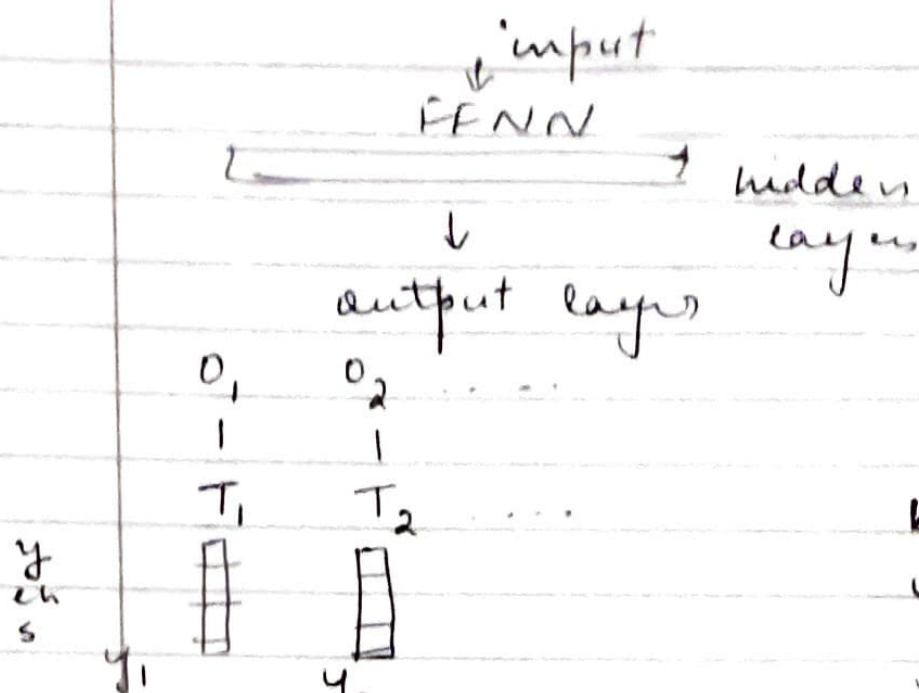
$\rightarrow$ we need to get the current seq. of phonemes at output level.

$\hookrightarrow$ this became the new obj. funcⁿ for speech researchers.

L8.6 Intro to CTC - 2



MFCC $O_1 O_2 O_3 O_4$
(80x)



but in CTC,
we have to
just align the
path of y s.

some non 0
values will be present
(they should add up to 1)

- in CTC, we take all possible paths, such that seq. order is maintained.

Try to maximize the sum of all path probabilities that ensures the correct seq.

3 states - no. of times 3^L paths
 $L \rightarrow R$ } so this will also reduce the paths

- extend the CTC idea to characters.

→ get the correct seq. of characters.
since, lexicon is easier for phonemic lang.

- this is also known as end-to-end speech recogniⁿ. Now, we don't need a lexicon.